

CERTIFIED BELLMAN OPERATORS FOR COSTLY POLICY REVISION

QIAN QI
Peking University

An organization revises a decision rule at minimum implementation cost while protecting operating performance after every state–date restart. One randomized Markov continuation must serve all restarts. We characterize the occupation relaxation optimized by restart prices and certify the loss from a computed price portfolio. Refinement over shared regret budgets restores the common continuation with an explicit approximation bound. For continuous-density kernels, restriction and averaging yield paired finite approximations of the original Borel-policy optimum. Uniform-reset dynamics admit a linear moment representation and integrated lower certificates valid for all Borel policies. In a source-frozen study, all 39 diffuse-state models attain certificate width 10^{-3} with each of two proposal backends. Separate occupation audits tightly resolve the price optimum in six previously unfinished finite cases. All historical intervals and unsuccessful finite refinements are retained.

KEYWORDS: Dynamic programming, Policy revision, Common Markov policies, Constrained control, Verified computation.

1. INTRODUCTION

An organization may promise operating protection whenever its decision rule is restarted while evaluating implementation expense under its initial population. These commitments restrict a single policy in different ways. The operating promise applies at every state and date, including states that have little initial probability. Implementation expense is an *ex ante* objective and is affected by the states the policy subsequently visits. Plans optimized separately for individual restarts need not prescribe the same behavior at their common successor states.

We study minimum-cost policy revision under this common-continuation requirement. The decision variable is a Borel randomized Markov policy: one probability vector at each state and date, with fresh randomization conditional on that state and date. We do not replace it by a hidden initial lottery over persistent plans. The implementation cost can price departures from an installed action rule; operating protection requires that the value loss relative to operating-optimal behavior never exceed a specified allowance. The economic object is the least cost of meeting that promise, not the least initial-average operating loss.

The paper supplies a chain from this policy object to paired, independently readable cost bounds. Its first component identifies what restart prices optimize. An occupation-flow relaxation permits different incoming edges to allocate a successor’s regret allowance differently. A shared successor-budget equation restores the original policy problem, whereas the relaxed price problem can have a strict gap. A two-period rational example separates a constant multiplier, the complete restart-price family, and the common-policy optimum. We also construct feasible occupation witnesses that place an upper bound on the optimum of the *entire* price family. Thus a small numerical price bound is no longer confused with a certificate that price optimization has been exhausted.

The second component exploits the missing consistency directly. Instead of subdividing every action probability, a covering algorithm subdivides one regret budget for each noninitial, nonterminal state–date pair. Conditional on budget intervals, its lower problem is a sequence of one-constraint simplex programs. A common-policy repair supplies an upper endpoint. The approximation bound separates budget diameter, operating-witness error, and deployment

rounding. The coordinate count is $n(T - 1)$ rather than nTm ; the covering guarantee remains exponential. This is a structural reduction with a quantitative guarantee, not an inference of polynomial complexity from a bilinear reformulation.

The third component establishes paired approximation for genuinely diffuse states. For a kernel with a certified total-variation approximation by a cellwise density, averaging an arbitrary Borel policy gives a finite policy in the approximating model without changing its initial cost or violating its all-restart constraints. Uniform primitive-error bounds then transfer a finite lower endpoint at a relaxed allowance and a finite upper endpoint at a tightened allowance to the original continuous-state problem. The allowance perturbation vanishes under refinement by a constructive repair argument. The initial distribution is kept fixed; it is not replaced by a finitely observed population.

A uniform-reset subclass makes the continuum construction especially transparent. Its common continuation is represented by cumulative regret moments. The policy and the moments obey linear constraints, and a Lagrange certificate integrates the minimum of finitely many affine functions. This lower bound is valid against *all* Borel policies, not only the cell policies used to construct the upper bound. Endpoint checks establish operating protection everywhere on each cell. The associated numerical study varies horizon, action count, discounting, allowance, and spatial slope one factor at a time, with three new seeds. All 39 models attain the 10^{-3} absolute certificate target with both proposal backends. The kernels have continuous densities and the initial law is uniform; these results are distinct from the earlier finite-fleet theorem.

The finite comparisons identify a different computational boundary. At construction budgets of two and eight seconds, the new budget cover certifies 2 and 2 of the twelve historical primary cases, respectively; the probability-space price formulation certifies 4 and 4. We retain these outcomes rather than present a coordinate reduction as a universal speed improvement. A separate audit brackets the uncapped price optimum to width below 10^{-8} in six of the eight historically unclosed cases. Further price searching alone cannot resolve a common-policy gap in those rows. The other two price audits remain visibly wider. The thirty earlier positive-cost intervals for moving atomic dynamics under a uniform initial law are retained unchanged; the new density theorem is not invoked across their singular transition approximation.

1.1. Relation to the literature

Occupation measures are a standard representation for expected constrained Markov decision problems (Altman, 1999). Theorem 1 instead derives the occupation program of this particular restart-price family and identifies its missing common-budget equations. The strict separation in Theorem 2 is essential: ordinary occupation duality does not turn an initial-average constraint into every restart’s constraint. Information-relaxation bounds (Brown, Smith, and Sun, 2010, Brown and Smith, 2014) relax information restrictions; our price relaxation separates allocations of a common successor budget and does not change nonanticipativity.

The LP approximation of a value function in a chosen basis (de Farias and Van Roy, 2003) and finite-state approximation of constrained control (Dufour and Prieto-Rumeau, 2013, Saldi, 2018) address related numerical problems. The distinctive transfer in Theorem 6 concerns both endpoints of an all-state constrained *revision-cost* optimum, with the same initial law and an explicit allowance correction. Its averaging lemma is stated for the whole Borel policy class. Weak convergence alone would not control arbitrary discontinuous policies; the total-variation assumption is deliberate. The existing exact finite-atomic restriction–extension theorem is retained as a separate result.

Robust dynamic programming uses uncertainty sets for transition laws and rectangularity to obtain recursive optimization (Iyengar, 2005, Nilim and El Ghaoui, 2005). We keep the target

transition law fixed and use an enclosure of its approximation error to protect the target policy, rather than optimize a newly specified robust-control objective. Safe policy improvement from batch data protects performance relative to a data-generating baseline (Laroche, Trichelair, and Tachet des Combes, 2019). Here the operating reference is the optimal operating value, the guarantee is at every restart, and implementation cost is minimized subject to that guarantee. Statistical kernel uncertainty would require an additional simultaneous error event; arithmetic certification alone is not such an event.

The covering constructions use established global-optimization ideas, including McCormick envelopes (McCormick, 1976). Moment hierarchies offer another approach to polynomial optimization (Lasserre, 2001); we do not claim dominance over them. Our contribution is a smaller continuation-budget cover, its explicit repair modulus, and an auditable cost interval for the original policy object. The reset reduction similarly uses linear semi-infinite optimization principles (Hettich and Kortanek, 1993), but provides explicit cumulative-moment equations, a Borel-valid integrated dual, and a whole-cell deployment check for this contract. Numerical proposals are separated from mathematical evidence as in verified computation (Rump, 2010).

The economic motivation is computational rather than an estimated application. Dynamic economic modeling requires approximation and error accounting (Judd, 1998, Brumm and Scheidegger, 2017); an operating oracle with certified error can enter our bounds without revealing exact values. The present experiments do not establish the cost of a learned high-dimensional oracle. Unlike the replacement application of Rust (1987), the fleet and reset primitives here are designed, not estimated. Decision-cost normalization is therefore reported without an empirical dollar or welfare interpretation.

2. THE COMMON-POLICY REVISION PROBLEM

Let X be a standard Borel space, $A = \{1, \dots, m\}$, and $t = 0, \dots, T-1$. The known transition kernels P_t^a are Borel. Rewards, nonnegative implementation costs, and terminal rewards are Borel and satisfy

$$|r_t(x, a)| \leq R, \quad 0 \leq k_t(x, a) \leq K, \quad |g(x)| \leq G.$$

Throughout, $0 < \beta < 1$, $T < \infty$, and $\varepsilon > 0$. Define $S_h = \sum_{j=0}^{h-1} \beta^j$ and $S_0 = 0$. A policy p is a family of Borel probability vectors $p_t(\cdot | x)$. Its operating and implementation values and the optimal operating value satisfy

$$\begin{aligned} J_T^p &= V_T = g, & C_T^p &= 0, \\ V_t(x) &= \max_a \{r_t(x, a) + \beta P_t^a V_{t+1}(x)\}, \end{aligned} \tag{1}$$

$$J_t^p(x) = \sum_a p_t(a | x) \{r_t(x, a) + \beta P_t^a J_{t+1}^p(x)\}, \tag{2}$$

$$C_t^p(x) = \sum_a p_t(a | x) \{k_t(x, a) + \beta P_t^a C_{t+1}^p(x)\}.$$

For a specified initial probability law ν , the target is

$$K_R(\nu, \varepsilon) = \inf_{p \in \mathcal{F}(\varepsilon)} \int C_0^p d\nu, \quad \mathcal{F}(\varepsilon) = \{p : J_t^p(x) \geq V_t(x) - \varepsilon \text{ for every } t, x\}. \tag{3}$$

An operating-optimal Borel policy exists by backward first-maximizer selection, so this feasible set is nonempty. We use an infimum when a general Borel optimizer has not been established.

Finite models have a compact policy simplex and continuous constraints, hence attain the minimum.

Set

$$d_t(x, a) = V_t(x) - r_t(x, a) - \beta P_t^a V_{t+1}(x), \quad D_t^p = V_t - J_t^p. \quad (4)$$

Then $d_t \geq 0$, $D_T^p = 0$, and $D_t^p = \sum_a p_t(a | x)(d_t(x, a) + \beta P_t^a D_{t+1}^p(x))$. The economic restriction is $0 \leq D_t^p \leq \varepsilon$ everywhere. Its initial-law analogue, $\int D_0^p d\nu \leq \varepsilon$, is weaker and will be used only as a comparison.

For an installed rule a^0 , a representative cost is $k_t(x, a) = w_t(x) \mathbf{1}\{a \neq a_t^0(x)\}$. Physical switching from the previous action is a different friction and requires augmenting the state. A fixed lottery-administration fee f can be compared with a certificate in a designed budget exercise, but is not silently included in k . Scaling implementation costs by $c > 0$ scales K_R, L, U , and an appropriate absolute target by c , while leaving the operating constraint unchanged.

3. WHAT RESTART PRICES OPTIMIZE

For bounded nonnegative Borel prices λ_t with $\lambda_T = 0$, define $u_T = 0$ and

$$u_t(x) = \min_a \{k_t(x, a) + \lambda_t(x)(d_t(x, a) - \varepsilon) + \beta P_t^a[u_{t+1} + \varepsilon \min\{\lambda_t(x), \lambda_{t+1}\}](x)\}. \quad (5)$$

Every feasible common policy satisfies $C_t^p + \lambda_t(D_t^p - \varepsilon) \geq u_t$. Indeed, backward substitution leaves the nonnegative hinge

$$(\lambda_t(x) - \lambda_{t+1}(y))_+ D_{t+1}^p(y) + (\lambda_{t+1}(y) - \lambda_t(x))_+ (\varepsilon - D_{t+1}^p(y)).$$

Together with $C_t^p \geq 0$, this proves the lower bound

$$L(\lambda) = \int \max\{0, u_0\} d\nu \leq K_R(\nu, \varepsilon). \quad (6)$$

The same induction permits lower disadvantages in place of d , an observation used by the implemented witness-only method.

3.1. An exact finite occupation interpretation

For this subsection let $X = \{1, \dots, n\}$. Write $P_{tia,j}$ for the transition probability and let μ be any nonnegative initial subprobability. For nonnegative variables y_{tia} define $z_{ti} = \sum_a y_{tia}$ and $f_{tij} = \beta \sum_a y_{tia} P_{tia,j}$. Edges only run from date t to date $t + 1$, for $t < T - 1$. Set missing edge sums to zero. Consider the linear program

$$\mathcal{O}(\mu) = \min_{y, \alpha, \gamma \geq 0} \sum_{t,i,a} k_{tia} y_{tia}, \quad (7)$$

$$z_{0i} = \mu_i, \quad z_{ti} = \sum_h f_{t-1,hi} \quad (t \geq 1),$$

$$\sum_a y_{tia} (\varepsilon - d_{tia}) \geq \sum_j \alpha_{tij} + \sum_h \gamma_{t-1,hi}, \quad (8)$$

$$\alpha_{tij} + \gamma_{tij} \geq \varepsilon f_{tij}. \quad (9)$$

The variables y are discounted occupations. The two nonnegative allocations on an incoming edge need not encode the same successor regret as allocations from other incoming states.

THEOREM 1—Price supremum and common-budget completion: *For every finite model satisfying Section 2,*

$$\sup_{\lambda \geq 0} \sum_i \mu_i u_{0i}(\lambda) = \mathcal{O}(\mu). \quad (10)$$

Both sides have finite attained optima. Adding variables $b_{ti} \in [0, \varepsilon]$, $1 \leq t < T$, and the equations

$$\alpha_{tij} = f_{tij} b_{t+1,j}, \quad \gamma_{tij} = f_{tij}(\varepsilon - b_{t+1,j}) \quad (11)$$

to (7)–(9) makes its optimum equal to the original all-restart common-policy value with objective measure μ . Thus price optimization solves an explicitly identified relaxation; restoring shared successor budgets solves the stated policy problem.

PROOF: Introduce free variables v_{ti} and nonnegative variables λ_{ti}, h_{tij} , with $v_T = \lambda_T = 0$. The price linear program maximizes $\sum_i \mu_i v_{0i}$ subject to

$$v_{ti} - \beta \sum_j P_{tia,j} v_{t+1,j} + (\varepsilon - d_{tia}) \lambda_{ti} - \beta \sum_j P_{tia,j} h_{tij} \leq k_{tia}, \quad (12)$$

$$h_{tij} \leq \lambda_{ti}, \quad h_{tij} \leq \lambda_{t+1,j}.$$

There are no h variables in the last period. Given a price field, increasing h to the corresponding minimum can only relax these inequalities. Backward maximization of v then gives exactly (5). Conversely, (5) and those minima are feasible. This identifies the linear-program value with the supremum on the left of (10).

Assign dual multipliers y to (12) and α, γ to the two upper bounds on h . The free v variables give occupation conservation. The nonnegative λ variables give (8); the nonnegative h variables give (9). The resulting dual is precisely (7). It is feasible: for any feasible policy set $y_{tia} = \beta^t \Pr_\mu^p(X_t = i) p_t(a | i)$, $\alpha_{tij} = f_{tij} D_{t+1,j}^p$, and $\gamma_{tij} = f_{tij}(\varepsilon - D_{t+1,j}^p)$. At a positive-date node the budget inequality is an equality; at a root it is its all-restart regret inequality. The primal price program is also feasible, for example with zero prices and sufficiently small v . Its value is bounded by a feasible policy's cost. Finite-dimensional linear-program duality proves equality and attainment.

The same policy construction satisfies (11) with $b = D^p$, so the completed program has value at most the original optimum. In the reverse direction, take a completed feasible solution. At every node with $z_{ti} > 0$, put $p_t(a | i) = y_{tia} / z_{ti}$. Conservation ensures that a positive-probability transition of this policy goes to a positive-occupation node. For $t \geq 1$, substituting (11) in (8) and using incoming conservation gives

$$b_{ti} \geq \sum_a p_t(a | i) \{d_{tia} + \beta \sum_j P_{tia,j} b_{t+1,j}\},$$

with terminal budget zero. At roots of positive μ mass the same argument replaces b_{0i} by ε . Backward induction gives actual regret at most b at positive-date occupied nodes and at most ε at roots. At an unoccupied node choose a first maximizer of $r_t + \beta P_t J_{t+1}^p$ after the later policy has been completed. If the future policy is globally feasible, this yields $J_t^p \geq V_t - \beta \varepsilon \geq V_t - \varepsilon$. Occupied transitions do not enter unoccupied successors, so this extension does not change their values. It is globally feasible and has objective $\sum k y$. This proves the reverse inequality. *Q.E.D.*

The theorem does not say that a usual occupation-measure formulation is incapable of expressing common policies. It gives the additional bilinear compatibility which this particular

price relaxation omits. The problem-specific content is the precise edge allocation and its completion, not linear-program duality itself.

COROLLARY 1—Clipping and constant multipliers: *In a finite model,*

$$\sup_{\lambda} L(\lambda) = \max_{S \subseteq \text{supp } \nu} \mathcal{O}(\nu \mathbf{1}_S). \quad (13)$$

For a constant price $\lambda_{ti} = \ell$ at all nonterminal dates, the un-clipped objective is

$$\sum_i \nu_i u_{0i} = \inf_p \left\{ \int C_0^p d\nu + \ell \left(\int D_0^p d\nu - \varepsilon \right) \right\}.$$

Consequently the best constant-multiplier bound is no larger than the un-clipped state–date–price supremum, which is no larger than K_R .

PROOF: For every vector v , $\sum_i \nu_i \max(0, v_i) = \max_S \sum_{i \in S} \nu_i v_i$. A finite maximum commutes with a supremum, so Theorem 1 proves (13). For the second statement, the price correction cancels successive constant-price allowance terms. Summing them along a trajectory leaves $-\ell\varepsilon$, while the disadvantages telescope to D_0^p . Backward minimization of the transformed objective proves the identity. Q.E.D.

The implementation uses a finite price cap and numerically proposed prices. It enumerates all nonempty masks when $n \leq 4$ and otherwise proposes the full mask and positive-mass singletons, subject to a declared time budget. Every proposed field is reconstructed rationally and ranked by its clipped certificate. A truncated mask search is not advertised as attaining (13); the earlier un-clipped-only objective is not silently equated with the displayed bound. Appendix A.2 gives an exact example in which the full-mask optimum is $3/8$ while a clipped field attains $15/32$.

3.2. A strict separation with exact rational values

THEOREM 2: *There is a two-state, two-period, two-action model with action-independent transitions for which the constant-multiplier optimum, the clipped restart-price supremum, and the common-policy optimum are respectively*

$$\frac{5}{4} < \frac{27}{16} < \frac{7}{4}. \quad (14)$$

The first two values have exact matching primal and dual witnesses. The final value follows from a one-variable affine minimization.

PROOF: Take $\beta = \varepsilon = 1/2$, zero terminal reward, and actions G, B with operating rewards $0, -1$. Every transition goes to state 1. At date 0 the cost of G is 1 in state 1 and 3 in state 2; at date 1 it is $3/2$ in either state. Action B costs zero. Let $\nu = (1/2, 1/2)$. Then $V = 0$ and $d(G) = 0, d(B) = 1$.

Let q denote the probability of B at date 1 in state 1 and q_i the two date-0 probabilities. Feasibility requires $0 \leq q \leq 1/2$ and $q_i \leq 1/2 - q/2$. Costs decrease with each q_i , so they equal these upper limits at an optimum. The objective becomes $7/4 + q/4$, minimized at $q = 0$ with value $7/4$.

Set $\lambda_0 = (1, 3)$ and $\lambda_1 = (3/2, 3/2)$. The recurrence gives $u_1 = (3/4, 3/4)$ and $u_0 = (9/8, 9/4)$, hence the lower value $27/16$. A feasible occupation witness in (7) takes the date-0

probabilities of B as $(1/4, 1/2)$ and the date-1 probability as $1/4$. The discounted occupations are $y_{01} = (3/8, 1/8)$, $y_{02} = (1/4, 1/4)$, and $y_{11} = (3/8, 1/8)$. On the two incoming edges use $(\alpha, \gamma) = (1/8, 0)$ and $(0, 1/8)$. Every occupation, node-budget, and edge inequality holds, and the objective is $27/16$. These matching witnesses prove the un-clipped supremum.

Clipping cannot improve it. With initial mass only at state 1 the original optimum is $9/8$ (choose $q = 1/2$); with mass only at state 2 it is $9/4$ (choose $q = 0$). Every price potential at the corresponding root is bounded by that positive optimum, hence so is its positive part. The same price field attains both bounds simultaneously.

Finally, for the expected-initial-regret problem choose $(q_1, q_2, q) = (0, 1, 0)$. Its average regret is $1/2$ and its cost is $5/4$. The constant price $\ell = 3$ gives $u_0 = (1/4, 9/4)$ and the same value. Weak duality proves the constant-multiplier optimum $5/4$. The conflicting preferred terminal randomizations explain the gap between the middle and final values: incoming edges cannot deploy different values of q at their common successor. *Q.E.D.*

The separation supplies a constructive reason to refine shared policy variables rather than assuming that more price proposals alone must close every interval. It also distinguishes a policy-to-certificate identity from a true optimization error. The former's clipping correction is explicitly negative: if its un-clipped nonnegative slack sum is S , then $C^p(\nu) - L(\lambda) = S - \int (-u_0)_+ d\nu$. We do not call this clipped expression a sum of nonnegative terms. The full earlier identity and its case-level diagnostics are retained in the supplement.

4. CERTIFYING THE PRICE-SEARCH LOSS

The price supremum is a relaxation value, but even that value need not be attained by a capped, time-limited, partially enumerated search. A certificate for the relaxation itself separates this numerical question from the common-policy question. Write $\mathcal{O}(\mu)$ for the occupation program in Theorem 1, and let $I = \{i : \nu_i > 0\}$. For $S \subseteq I$, put $\mu_i^S = \nu_i \mathbf{1}\{i \in S\}$. The clipped price optimum is

$$L_{\text{price}}^* = \max_{S \subseteq I} \mathcal{O}(\mu^S).$$

This is a finite maximum; states of zero initial weight need not be enumerated in the masks.

PROPOSITION 1—A complete portfolio-loss certificate: *For every nonempty initial-support mask S , suppose B_S is the objective of a verified feasible occupation witness for $\mathcal{O}(\mu^S)$, or the masked cost of a verified feasible common policy. Let ℓ be any clipped price lower endpoint reconstructed from nonnegative prices and valid operating lower disadvantages. Then*

$$\ell \leq L_{\text{price}}^* \leq B := \max(0, \max_{S \neq \emptyset} B_S), \quad 0 \leq L_{\text{price}}^* - \ell \leq B - \ell. \quad (15)$$

The upper bound is independent of the cap, candidate mask selection, and numerical time limit used to construct ℓ . A mask can be discarded from further optimization once its verified upper bound is below the best retained lower bound. A missing or unsuccessful occupation construction can always be replaced by the corresponding feasible-policy cost.

PROOF: Weak duality gives $\mathcal{O}(\mu^S) \leq B_S$. The common-policy completion in Theorem 1 shows that a feasible policy supplies an occupation witness for every masked initial measure, so the alternative upper bound is valid. Taking the maximum over masks gives the right inequality. The price inequality, witness monotonicity, and mask characterization give the left inequality. No numerical claim of LP optimality is used. The pruning assertion follows immediately from the two inequalities. *Q.E.D.*

We construct an occupation witness without trusting approximate flow equalities. Normalize a numerical occupation row into an action distribution and propagate that distribution from μ^S using exact transitions. The resulting y satisfies flow conservation exactly. On each edge put $f_{tij} = \beta \sum_a y_{tia} P_{tiaj}$. Reconstruct a number $b_{tij} \in [0, \varepsilon]$ from the numerical allocation and set

$$\alpha_{tij} = f_{tij} b_{tij}, \quad \gamma_{tij} = f_{tij} (\varepsilon - b_{tij}).$$

These equations enforce the edge inequalities with equality, but the state capacities may still fail. The following repair is entirely in the occupation relaxation; it does not pretend that its edge budgets already define a common policy.

LEMMA 1—Strict occupation repair: *In a finite model with exact disadvantages d , let $\bar{d} = \max d_{tia}$. At every row mix an operating-best action with the uniform action distribution, giving the uniform distribution weight*

$$\theta = \min\{1/2, \varepsilon(1 - \beta)/(4\bar{d})\}$$

when $\bar{d} > 0$, and $\theta = 1/2$ otherwise. Propagate its exact occupation y° from μ^S and set $\alpha^\circ = \gamma^\circ = \varepsilon f^\circ/2$. Its state-capacity slack is strictly positive at every state reachable from S under any action sequence. If s_{ti} is the capacity slack of a flow-feasible reconstructed candidate and s_{ti}° that of this reference, then mixing the two occupations with reference weight

$$\tau = \max_{s_{ti}^\circ < 0} \frac{-s_{ti}}{s_{ti}^\circ - s_{ti}}, \quad \max \emptyset = 0, \quad (16)$$

produces a feasible occupation witness. All quantities are computable by finite rational operations.

PROOF: At a noninitial state with mass z_{ti}° , the incoming γ° sum is $\varepsilon z_{ti}^\circ/2$, and the outgoing α° sum is at most $\beta \varepsilon z_{ti}^\circ/2$. The expected disadvantage is at most $\theta \bar{d} \leq \varepsilon(1 - \beta)/4$. Thus the capacity slack is at least $\varepsilon(1 - \beta)z_{ti}^\circ/4$. An initial state has no incoming allocation and an at least as large margin. At the terminal decision date there is no outgoing allocation, which also only increases the margin. The reference uses every action with positive probability, so its mass is positive wherever any candidate can send flow from S . At unreachable states both occupations and their edge allocations vanish. Flow conservation and the edge inequalities are linear and survive mixing. Equation (16) makes each remaining capacity slack nonnegative. *Q.E.D.*

An alternative finite completion uses only rational LP theory. After splitting free variables and adding slacks, clear row denominators in the price program to obtain $Ax = b$, $x \geq 0$. Delete redundant equations and let r be the row rank and $H \geq 1$ an integer bound on the absolute entries of $[A \ b]$. For every mask the finite optimum has a basic feasible solution. Cramer's rule gives a common sufficient coordinate cap

$$\Lambda_* = r!H^r.$$

Indeed a nonsingular integer basis has determinant of magnitude at least one and every numerator determinant is bounded by $r!H^r$. The constraint matrix and right-hand side do not change with the mask. Enumerating masks and exact bases therefore completes the price problem in finitely many steps. This cap can be extremely large; we use occupation witnesses, not this worst-case number, in the numerical audit.

Proposition 1 is a bound on the loss of the *whole computed portfolio*. It does not assign an unidentified portion of that loss separately to cap choice, omitted masks, operating enclosures, and premature LP stopping. The fixed-field operating-enclosure loss is bounded separately in the appendix. When $B - \ell$ is small but the original common-policy interval remains wide, further searching within the price family cannot supply the missing common-continuation consistency.

5. END-TO-END CERTIFICATION FROM OPERATING ENCLOSURES

The algorithm's inputs are the rational finite model, ε , an interval target $\eta > 0$, and a work budget. The only operating information supplied to price generation, candidate construction, or node relaxations is a pair of certified Bellman witnesses. Exact optimal operating values and inherited feasible policies are not inputs.

5.1. Enclosures and feasible policy generation

Let $\underline{V}, \bar{V}$ satisfy terminal bounds and the componentwise Bellman inequalities

$$\underline{V}_T \leq g \leq \bar{V}_T, \quad \underline{V}_t \leq \max_a (r_t^a + \beta P_t^a \underline{V}_{t+1}), \quad \bar{V}_t \geq \max_a (r_t^a + \beta P_t^a \bar{V}_{t+1}). \quad (17)$$

Backward induction gives $\underline{V} \leq V \leq \bar{V}$. Write $w_t = \|\bar{V}_t - \underline{V}_t\|_\infty$ and $\xi = \max_{t < T} w_t$. The implementation uses exact rational sums followed by outward dyadic rounding in each Bellman pass; it never computes an exact optimal V as an intermediate oracle. In representable tied examples, the two witnesses may coincide, which is disclosed rather than described as difficult approximation.

The two useful advantage quantities are

$$\underline{d}_{tia} = \max\{0, \underline{V}_{ti} - r_{tia} - \beta P_{tia} \bar{V}_{t+1}\} \leq d_{tia}, \quad (18)$$

$$d_{tia}^0 = \underline{V}_{ti} - r_{tia} - \beta P_{tia} \underline{V}_{t+1}. \quad (19)$$

The second need not be nonnegative. For $B_t^p = \underline{V}_t - J_t^p$, it gives the exact signed recursion $B_t^p = \sum_a p_{tia} (d_{tia}^0 + \beta P_{tia} B_{t+1}^p)$, with the appropriate actual terminal payoff. The lower-node program uses $\underline{V} - \varepsilon \leq J \leq \bar{V}$, not a fiction that $\underline{V}$ is an exact optimal value.

To create the initial incumbent, proceed backwards. Given the already constructed continuation, put

$$Q_a = r_t(x, a) + \beta P_t^a \hat{J}_{t+1}(x), \quad Z_a = k_t(x, a) + \beta P_t^a \hat{C}_{t+1}(x).$$

Minimize $\sum_a p_a Z_a$ over the probability simplex subject to $\sum_a p_a Q_a \geq \bar{V}_t(x) - \varepsilon$. An extreme minimizer has at most two positive actions and is found by enumerating feasible pure actions and pairs straddling the threshold. This is a conditional greedy construction, not a claim of globally optimal implementation. It supplies a feasible policy without any earlier run. A separately timed local nonlinear proposal can improve it; every resulting candidate is repaired and checked exactly.

LEMMA 2—Upper-witness repair: *Put $s = (1 - \beta)\varepsilon$ and assume $\xi < s$. If a common proposal q satisfies $J_t^q \geq V_t - \varepsilon - \delta$ everywhere, a backward repair produces $\hat{p}$ with $J_t^{\hat{p}} \geq \bar{V}_t - \varepsilon$ and $J_t^{\hat{p}} \geq J_t^q$. Its maximum row total-variation displacement, before deployment quantization, is at most*

$$\min \left\{ 1, \frac{\delta + \xi}{\delta + s} \right\}. \quad (20)$$

Rounding every action except an operating-best continuation action down to a mesh h and assigning the remaining probability to that best action preserves feasibility and adds at most $(m-1)h$ to this bound.

PROOF: At a given step evaluate the original row of q against the repaired future. By induction its value v is no smaller than J_t^q . The best continuation action has value $M \geq V_t - \beta\varepsilon$. Its margin above $\bar{V}_t - \varepsilon$ is at least $s - \xi > 0$. If v is below the threshold, mix the original row with that best action using the minimal fraction. If the threshold deficit is z , then $z \leq \delta + \xi$ and the fraction is at most $z/(z + s - \xi) \leq (\delta + \xi)/(\delta + s)$. Otherwise no mixture is needed. Values weakly increase at each step. Downward rounding of nonbest probabilities moves at most $(m-1)h$ probability mass to a weakly better action, so preserves both value inequalities. *Q.E.D.*

For policies whose rows differ by at most ρ in total variation, uniform perturbation constants sufficient here are

$$D_J = 2(RS_T + G)S_T, \quad D_C = K \sum_{t=0}^{T-1} \beta^t S_{T-t}. \quad (21)$$

Then $\|J^p - J^q\|_\infty \leq D_J \rho$ and $|C^p(\nu) - C^q(\nu)| \leq D_C \rho$. These conservative constants follow by bounding the range of each continuation action value and iterating the one-step perturbation inequality. They require no action separation.

5.2. Two node formulations and exact endpoints

A node is a box for all nTm probabilities, intersected rowwise with the simplex. At a nonempty box, rectangular Bellman minimization and maximization give exact bounds $J^{\min}, J^{\max}, C^{\min}, C^{\max}$ for every policy in the box. Each extremum is itself attained by a common Markov selector because the box is rectangular across state-date rows. A node can be eliminated if its simplex intersection is empty or if $J_{ti}^{\max} < \underline{V}_{ti} - \varepsilon$ somewhere. These are rational conditions, not numerical infeasibility flags.

The direct-value formulation writes the original Bellman equations in J, C and uses McCormick envelopes for each aggregate product $p_{tia}(r_{tia} + \beta P_{tia} J_{t+1})$ and its implementation analogue. The price formulation writes the signed $B = \underline{V} - J$ equations and disaggregates products by successor. It imposes $\sum_a (p_{tia} B_{t+1,j}) = B_{t+1,j}$ and the analogous cost identity. It additionally uses the certificate planes

$$C_{ti} + \lambda_{ti}\{B_{ti} + w_{ti} - \varepsilon\} \geq u_{ti}, \quad C_{ti} \geq \max\{0, u_{ti}\}, \quad (22)$$

where $w_{ti} = \bar{V}_{ti} - \underline{V}_{ti}$ and u is reconstructed with (18). Since $D_{ti}^p \leq B_{ti}^p + w_{ti}$, the first plane follows from the price inequality for every truly feasible policy. The variable shift and product identities do not change the underlying common-policy object. Neither formulation dominates the other as a general polyhedral statement.

For a node relaxation $\min c'z$ subject to $Az = b$, $Ez \leq f$, and $l \leq z \leq u$, rounded proposals y and $\tau \geq 0$ give the exact rational lower endpoint

$$y'b - \tau'f + \sum_j \min\{r_j l_j, r_j u_j\}, \quad r = c - A'y + E'\tau. \quad (23)$$

No dual feasibility tolerance is needed: the residual is explicitly paid. The node lower bound is the maximum of this endpoint, zero, the rectangular cost bound, the reconstructed price bound,

and its inherited parent bound. An exact common feasible policy supplies the upper endpoint. The global lower bound is the minimum over the complete leaf cover, clipped above by that feasible upper value. The checker reconstructs each row and split independently of the optimizer.

5.3. A matched finite stopping result

THEOREM 3—Witness-only finite completion: *Let a retained policy box have maximum row total-variation diameter at most ρ . Suppose (17) holds, $\xi < s$, and its rectangular operating test does not eliminate it. Evaluating and repairing any box member gives a cost at most*

$$\Omega(\rho, \xi, h) = D_C \left[\rho + \min \left\{ 1, \frac{D_J \rho + 2\xi}{D_J \rho + \xi + s} \right\} + (m-1)h \right] \quad (24)$$

above the box's rectangular cost lower bound. Consequently, choose witnesses and deployment mesh with

$$D_C \{ \xi/s + (m-1)h \} \leq \eta/64, \quad \xi < s. \quad (25)$$

Without work caps, midpoint best-bound search that bisects a widest coordinate at least every fourth depth, and evaluates a box member, terminates with a valid interval of width at most η . The statement holds for both node formulations, even when every numerical LP proposal is unavailable.

PROOF: A retained box has $J^{\max} \geq \underline{V} - \varepsilon$ everywhere. Any member q differs from its rectangular operating-maximizing policy by at most ρ per row, hence

$$J^q \geq V - \varepsilon - (\xi + D_J \rho).$$

Apply Lemma 2 with $\delta = \xi + D_J \rho$. The rectangular cost minimizer is another policy in the same box, so its objective differs from $C^q(\nu)$ by at most $D_C \rho$. The repair and deployment change the cost by at most D_C times their maximum row displacement. This proves (24). Relaxation strengthening and inherited lower bounds can only improve the interval.

Under (25), the limit of (24) as $\rho \downarrow 0$ is at most $2D_C \xi/s + D_C(m-1)h \leq \eta/32$. Choose a positive coordinate width w_* such that (24), with $\rho = mw_*$, is below η . There are $d = nTm$ coordinates. Along a path, after at most $4d \lceil \log_2(1/w_*) \rceil$ levels every coordinate has width at most w_* : the scheduled widest-coordinate splits alone suffice. Thus every deeper retained node already has a repaired feasible policy within η of its lower bound. While the global interval exceeds η , best-bound search selects a leaf below the current upper value minus η , which cannot be such a sufficiently refined node. Only finitely many shallower nodes exist. Exact elimination cannot remove a feasible solution. The asserted stopping follows. Q.E.D.

The node bound implicit in this proof is exponential in the sufficient depth. It is an accuracy guarantee, not a polynomial complexity claim. What changes relative to an idealized exact-oracle theorem is that the implemented pipeline supplies the witnesses and the feasible upper policy appearing in the proof, and its arithmetic level depends on the requested target.

For lower-price conditioning, the implementation starts at 8 bits and adds 8 bits until (25) holds and

$$\Lambda \sum_{t=0}^{T-1} \beta^t (w_t + \beta w_{t+1}) \leq \eta/64, \quad \Lambda = 4096. \quad (26)$$

The cap is fixed before model generation. Since the price field is generated only afterwards, this is an ex ante conditioning rule, not a favorable exact-model price reused with noisy values. Directed finite Bellman widths tend to zero as precision increases, so the rule terminates. Exact rational residuals still certify any proposed multipliers, irrespective of their quality. There is no claim that increasing precision alone eliminates optimization slack.

6. REFINEMENT OVER SHARED CONTINUATION BUDGETS

The exact occupation completion identifies a smaller set of coordinates in which to enforce consistency. In a finite model, give every state i at dates $t = 1, \dots, T - 1$ one interval $[l_{ti}, h_{ti}] \subseteq [0, \varepsilon]$ for its *actual cumulative regret*. Set $l_{0i} = 0$, $h_{0i} = \varepsilon$, and $l_{Ti} = h_{Ti} = 0$. One budget is shared across all incoming edges. There are $d_B = n(T - 1)$ variable coordinates, independently of the action count.

Let $\underline{d}$ be the nonnegative lower disadvantage constructed from the operating witnesses. At each row define the necessary local polytope

$$\mathcal{P}_{ti}(l, h) = \left\{ p \in \Delta_m : \sum_a p_a \left(\underline{d}_{tia} + \beta \sum_j P_{tiaj} l_{t+1,j} \right) \leq h_{ti} \right\}. \quad (27)$$

Starting from $H_T = 0$, compute

$$H_{ti} = \min_{p \in \mathcal{P}_{ti}(l, h)} \sum_a p_a \left(k_{tia} + \beta \sum_j P_{tiaj} H_{t+1,j} \right). \quad (28)$$

An empty row certifies an empty budget box. Otherwise each minimum is attained by a feasible pure action or a mixture of two actions. To see this, a simplex with one additional inequality has extreme points with at most two nonzero coordinates. Enumerating the m pure actions and the crossing pairs is therefore an exact rational node solver, without a floating LP or a bilinear product at that node.

PROPOSITION 2—Budget-box lower bound: *For any feasible common policy whose regrets lie in the specified box, $C_{ti}^p \geq H_{ti}$ at every state and date. Consequently νH_0 is a valid box lower bound. A partition of $[0, \varepsilon]^{d_B}$ by such boxes, together with a feasible policy of cost U , gives the global lower endpoint*

$$L = \min\{U, \min_{B \text{ live}} \nu H_0(B)\}.$$

Infeasible boxes may be discarded, and child lower bounds may be increased to their parent's lower bound.

PROOF: The regret recursion and $D_{t+1}^p \geq l_{t+1}$ imply that the row of any such policy lies in (27). Backward induction in the nonnegative transition probabilities proves $C^p \geq H$. Every feasible common policy has a regret vector in the original cube and therefore in a retained box or a certified infeasible box; the latter alternative is impossible. The global statement and inherited bounds follow. Q.E.D.

6.1. Price-enhanced budget nodes

A nonnegative price field can tighten a budget node without adding coordinates. Set $Z_T = 0$ and

$$Z_{ti} = \max \left\{ \lambda_{ti} l_{ti}, \min_{p \in \mathcal{P}_{ti}(l, h)} \sum_a p_a \left[k_{tia} + \lambda_{ti} \underline{d}_{tia} \right. \right. \\ \left. \left. + \beta \sum_j P_{tiaj} \{ Z_{t+1, j} + \min((\lambda_{ti} - \lambda_{t+1, j}) l_{t+1, j}, (\lambda_{ti} - \lambda_{t+1, j}) h_{t+1, j}) \} \right] \right\}. \quad (29)$$

Then $C_{ti}^p + \lambda_{ti} D_{ti}^p \geq Z_{ti}$ for every policy represented by the box. The first entry uses $C^p \geq 0$ and $D^p \geq l$. For the second, substitute the regret and cost recursions and bound $(\lambda_{ti} - \lambda_{t+1, j}) D_{t+1, j}^p$ over its interval. Therefore

$$\sum_i \nu_i \max\{0, Z_{0i} - \lambda_{0i} \varepsilon\}$$

is another node lower bound. We take the maximum of this bound, the cost-node bound, zero, and the parent bound. Prices strengthen a valid cover but are not required for its completion.

6.2. A computable completion modulus

We give a modulus for the cost-minimizing row policy in (28); retaining this policy is important even when a price-enhanced node has a different minimizer. Let $w = \max_{t, i} (h_{ti} - l_{ti})$ over the variable coordinates, and suppose

$$0 \leq d_{tia} - \underline{d}_{tia} \leq e_d, \quad 0 \leq \bar{V}_{ti} - V_{ti} \leq \xi, \quad \xi < s := (1 - \beta)\varepsilon.$$

All these bounds are obtained from the supplied operating enclosures. Define

$$\delta = (\beta w + e_d) S_T, \quad D_C = K \sum_{t=0}^{T-1} \beta^t S_{T-t}.$$

Here row total variation is half the ℓ^1 distance; the cost Lipschitz constant D_C is the same as in the finite refinement theorem.

THEOREM 4—Budget-cover completion: *Let q be the common Markov policy obtained from the minimizers of (28). Its exact cost is νH_0 and its all-state operating violation is at most δ . Backward repair against $\bar{V} - \varepsilon$, followed by downward quantization of the nonbest probabilities on a 2^{-b} grid, gives a feasible policy $\hat{q}$ with*

$$C_0^{\hat{q}}(\nu) - \nu H_0 \leq D_C \left\{ \frac{\delta + \xi}{\delta + s} + (m - 1) 2^{-b} \right\}. \quad (30)$$

Thus a complete cover with this expression at most η certifies the target width η . For fixed $\eta > 0$, sufficiently accurate witnesses, sufficiently fine deployment precision, and a sufficiently fine budget cover achieve the target after finitely many rational node operations.

PROOF: The Bellman minimizers use the same continuation H at every incoming edge, so evaluating their common policy gives exactly H . Write $E_t = \max_i (D_{ti}^q - h_{ti})_+$. From (27), $d \leq \underline{d} + e_d$, and $D_{t+1}^q \leq h_{t+1} + E_{t+1}$,

$$E_t \leq e_d + \beta w + \beta E_{t+1}, \quad E_T = 0.$$

The term βw can be omitted at the last decision date; retaining it gives $E_t \leq \delta$. In particular $D^q \leq \varepsilon + \delta$ everywhere. During backward repair the actual successor operating values only increase. If a row remains below $\bar{V}_t - \varepsilon$, its shortfall to that threshold is at most $\delta + \xi$. The continuation-best action has operating value at least $V_t - \beta\varepsilon$, hence margin at least $s - \xi$ above the threshold. Mixing reaches the threshold with mass at most

$$(\delta + \xi) / \{(\delta + \xi) + (s - \xi)\} = (\delta + \xi) / (\delta + s).$$

The repaired row moves by no more than this amount in total variation. Rounding every nonbest coordinate downward transfers the removed mass to that same best action, preserves feasibility, and adds at most $(m - 1)2^{-b}$ to the row movement. The policy-cost Lipschitz inequality gives (30). On every live box, a retained feasible candidate is therefore within that amount of the cost lower bound, and the enhanced lower bound is no weaker. Covering the compact budget cube by a finite mesh proves the last assertion. Q.E.D.

The proof provides an explicit sufficient mesh. After allocating rounding and witness error, choose w so that (30) is at most η . A regular cover needs at most

$$\left\lceil \frac{\varepsilon}{w} \right\rceil^{n(T-1)}$$

boxes. Each node uses $O(Tnm^2)$ comparisons after $O(Tn^2m)$ continuation products. Rational bit lengths and the number of boxes are still consequential. For $T = 1$, no continuation coordinates are needed and the row program directly solves the finite problem up to witness and deployment error.

The implementation uses midpoint bisection. Every fourth depth splits a widest coordinate, while other splits prioritize disagreement between a candidate's actual regret and the assigned budget. Along any sufficiently deep path the widest-coordinate steps force the diameter to zero. Both the unpriced cost-node minimizer and the price-node minimizer are repaired and considered as incumbents. The complete cover, including pruned and infeasible boxes, is serialized for an independently written reader. Time limits are soft: a started exact node operation and final serialization and verification finish before the corresponding budget check. The measured all-in cost, not the declared construction budget, is reported.

7. A PAIRED CONTROLLED-CONTINUUM CONSTRUCTION

Consider the same Borel problem (3), with transition kernels

$$P_t^a(x, \cdot) = \sum_{s=1}^S \omega_{tas}(x) \delta_{F_{tas}(x)}(\cdot), \quad \omega_{tas} \geq 0, \quad \sum_s \omega_{tas} = 1, \quad (31)$$

where the maps and weights are Borel. They may be action dependent, noninvertible, and state dependent. Let the *objective* initial law have finite support H_0 , and recursively set

$$H_{t+1} = \bigcup_{x \in H_t, a, s: \omega_{tas}(x) > 0} \{F_{tas}(x)\}. \quad (32)$$

Equal successors are identified, not copied into history-specific nodes. This is essential: two paths that reach the same state and date must use the same future policy.

THEOREM 5—Finite observation and global extension: *Under Section 2 and (31), with a finitely supported objective initial law, the unrestricted Borel common-policy value equals the finite common-policy value on the forward sets (32):*

$$K_R^X(\nu, \varepsilon) = K_R^H(\nu, \varepsilon). \quad (33)$$

Every feasible finite policy has an explicit Borel extension satisfying the operating constraint at every point of X , not just at the forward sets. Therefore a paired finite certificate $[L, U]$ is a paired certificate for the full-state problem with exactly the same endpoints. The construction uses at most $|H_0|(mS)^t$ states at date t and incurs zero spatial approximation error.

PROOF: All possible successors of a point of H_t belong to H_{t+1} , so the operating Bellman value on these sets equals the restriction of V^X by backward induction. Every globally feasible policy restricts to a feasible finite policy and has the same objective. This proves $K_R^H \leq K_R^X$.

For the reverse direction take any finite feasible policy p^H . Put $\hat{J}_T = g$. After defining the later extension, set

$$\hat{p}_t(\cdot | x) = \begin{cases} p_t^H(\cdot | x), & x \in H_t, \\ \delta_{a_t(x)}, & x \notin H_t, \end{cases} \quad a_t(x) = \text{first maximizer of } r_t(x, a) + \beta P_t^a \hat{J}_{t+1}(x). \quad (34)$$

Finite sets are Borel. Finite-action first-maximizer selection and the Borel kernels make the extension Borel. Values on H_t are exactly the finite values because all successors stay on the forward sets. They satisfy $\hat{J}_t \geq V_t - \varepsilon$. Off H_t , assuming global feasibility at date $t + 1$, one has

$$\hat{J}_t(x) = \max_a \{r_t(x, a) + \beta P_t^a \hat{J}_{t+1}(x)\} \geq V_t(x) - \beta\varepsilon \geq V_t(x) - \varepsilon.$$

This completes the induction. Starting under ν , the policy never leaves the finite forward sets, so its implementation objective is unchanged. Taking infima proves the reverse inequality. Each state has at most mS successors, which gives the size bound. No state or probability is perturbed, establishing the zero spatial error. Q.E.D.

COROLLARY 2—Executable paired accuracy: *Suppose the finite forward model can be constructed with exact rational arithmetic, including decidable equality of successor states. Apply Theorem 3 to that model and use (34) off the forward sets. This gives a computable paired hierarchy and a stopping rule $U - L \leq \eta$ for (3). A running maximum of certified lower endpoints and running minimum of feasible upper endpoints give monotone sequences converging to the same value when the work caps are removed and targets tend to zero.*

The extension is also operational for the affine rational model below. At any queried state, evaluate its finitely branching continuation and test membership in the finite rational forward sets. This computes an action without an exact global operating oracle. In a piecewise-affine implementation, ordinary max-affine breakpoints coexist with exceptional preimages of the assigned finite policies. Both counts may grow exponentially with the horizon. The theorem guarantees a representation and a matched certificate, not cheap symbolic compilation. The publication contract checker independently reconstructs all forward states, merges, transition rows, rewards, and the terminal folding used in the experiment.

7.1. An observed-fleet service contract with original controlled atoms

The continuous condition is $x \in [0, 1]$. Operating rewards are $r(x, a) = x - c_a$, with

$$(c_0, c_1, c_2) = (0, 3/20, 9/20), \quad g(x) = x/2, \quad \beta = 19/20.$$

An installed action 0 costs nothing to retain; either departure costs $1 + x$. The two shock outcomes for the three actions are

$$\begin{aligned} F_0(x) &\in \{3x/4, 3x/4 + 1/20\}, \\ F_1(x) &\in \{17x/20 + 1/10, 17x/20 + 7/50\}, \\ F_2(x) &\in \{x/10 + 4/5, x/10 + 17/20\}, \end{aligned}$$

with probabilities $3/5, 2/5$. These are the original affine controlled atoms, not a finite-state replacement for different dynamics. Condition affects reward, revision expense, and every controlled successor. We fix $\varepsilon = 1/100$ and the observed initial fleet law $\nu = (3/4)\delta_{1/5} + (1/4)\delta_{4/5}$, at horizons 2 and 3.

A service organization can interpret the statewise allowance as a warranty on remaining operating value whenever its rule is restarted, including off-fleet states; the objective weights the vehicles in the observed fleet. This is a specified decision contract, not evidence that any particular organization uses it. The extension in Theorem 5 is what makes the warranty global despite the finite objective population. Lottery administration is assumed permitted by that contract and is charged separately if nonzero.

The two certified optimal costs are approximately 1.2696 and 2.8687 normalized implementation units. Hence a two-unit budget permits the two-period task but not the three-period task under the stated operating guarantee. This is an internally verified budget distinction in designed primitives, not an estimate of market maintenance costs or empirical welfare. Multiplying the cost scale and budget by the same positive constant preserves the conclusion. The earlier arbitrary monetary conversion is retained only in the clearly identified historical manuscript.

For horizon 2 the active forward-set sizes are 2, 11, 63; for horizon 3 they are 2, 11, 63, 367. The last transition's expected g is folded into the last-period reward exactly, avoiding terminal padding. The policy problem has respectively 13 and 76 active decision nodes. Dummy nodes used for rectangular storage carry zero costs and have no incoming transitions from active states. Their equations are checked separately; they do not stand for omitted physical states. Table VI reports the paired intervals.

7.2. What is not an initial-law approximation

Equation (33) keeps ν fixed. It does not authorize replacing a uniform initial law by finitely many atoms. In particular, the earlier thirty positive-cost uniform-initial-law intervals are unchanged, and the earlier nonlinear two-state widths remain roughly 25–76% at allowance $1/2$. Their original results and proofs are reproduced in the retained technical material. The one-sided Borel density theorem likewise remains a one-sided theorem outside the hypotheses of Theorem 5. A general matched numerical hierarchy for diffuse initial laws with controlled moving atoms requires further structure and is not inferred from finite observation.

8. PAIRED APPROXIMATION WITH CONTINUOUS TRANSITION DENSITIES

Exact forward closure serves finitely supported initial populations under finite-atomic transitions. We now give a different route that leaves a diffuse initial law unchanged and permits

action-dependent continuous-density transitions. The approximation is two-sided for the original Borel-policy optimization, not merely an upper approximation within a grid policy class.

8.1. A cellwise-density model and unrestricted-policy averaging

Let X be compact and let $\{B_i\}_{i=1}^n$ be a finite Borel partition. Fix a probability measure μ with $\mu(B_i) > 0$, and write $\mu_i = \mu(\cdot \mid B_i)$. Consider a surrogate model whose rewards, costs, and terminal reward are constant on each cell, and whose kernels are

$$\bar{P}_t^a(x, dy) = \sum_j Q_{tiaj} \mu_j(dy), \quad x \in B_i, \quad Q_{tiaj} \geq 0, \quad \sum_j Q_{tiaj} = 1. \quad (35)$$

The coefficients can depend on the origin cell, action, and date. The within-destination-cell conditional law, however, is the same μ_j for every incoming edge. The surrogate remains a continuous-state process when μ is diffuse. Its finite model uses transitions Q and initial cell probabilities $\nu_i = \nu(B_i)$.

LEMMA 3—Exact averaging of Borel policies: *For every allowance $\epsilon > 0$, the surrogate's unrestricted Borel-policy revision optimum equals the finite cell model's common-policy optimum, with initial weights ν_i . A finite feasible policy extends as a constant policy on each cell. Conversely, every feasible Borel policy admits a feasible finite policy with exactly the same surrogate initial implementation cost. The original initial measure ν need not equal μ .*

PROOF: The surrogate optimal operating value is cellwise constant by backward induction. A cell-constant finite policy clearly has the same operating and implementation values in the two models. Conversely, take any Borel policy p . At dates $t \geq 1$, average its action probabilities on cell B_i with respect to μ_i . Suppose the resulting finite continuation values equal the μ_j -averages of the original continuation values. Equation (35) makes the continuation contribution at date t constant within each origin cell for each action. Averaging the policy-value identities therefore proves the same equality at date t . Backward induction begins at the cell-constant terminal reward. Averaging the original pointwise operating constraint proves the finite operating constraint at every such cell and date.

At date zero, average the policy with respect to $\nu(\cdot \mid B_i)$ whenever $\nu(B_i) > 0$, and with respect to μ_i otherwise. The date-one continuation still uses μ_j , so the same identities prove finite feasibility at date zero and equality of the initial implementation objective. Taking infima in both directions proves equality of optima. No history-dependent or persistent root lottery has been introduced; each averaged row is a single Markov action distribution. *Q.E.D.*

The common within-cell conditional law is the key assumption. A point-mass quadrature on a grid need not approximate a diffuse transition law in total variation and does not justify this lemma for the *target* model. We first obtain an exact finite representation of the surrogate, then account for its distance from the target model.

8.2. Uniform transfer with an allowance correction

Use $\text{TV}(P, Q) = \sup_A |P(A) - Q(A)|$. For a bounded function f , $|Pf - Qf| \leq \text{TV}(P, Q) \text{span}(f)$. Suppose certified primitive bounds are available:

$$\begin{aligned} \|r - \bar{r}\|_\infty &\leq a_r, & \|k - \bar{k}\|_\infty &\leq a_k, & \|g - \bar{g}\|_\infty &\leq a_g, \\ \sup_{t, x, a} \text{TV}(P_t^a(x, \cdot), \bar{P}_t^a(x, \cdot)) &\leq a_P. \end{aligned} \quad (36)$$

Let R_*, G_*, K_* bound both models' absolute rewards, absolute terminal rewards, and nonnegative costs. Define, backwards,

$$A_T = a_g, \quad A_t = a_r + \beta A_{t+1} + 2\beta a_P \{R_* S_{T-t-1} + \beta^{T-t-1} G_*\}, \quad (37)$$

$$E_T = 0, \quad E_t = a_k + \beta E_{t+1} + \beta a_P K_* S_{T-t-1}, \quad (38)$$

$$\zeta = 2 \max_{t < T} A_t.$$

These recursions bound errors uniformly over every Borel policy, including discontinuous policies. The same A_t also bounds the difference between the two optimal operating values.

THEOREM 6—Paired density approximation: *Assume (35)–(36) and $\varepsilon > \zeta$. Write $K_h(e)$ for the finite surrogate optimum at allowance e . Then*

$$K_h(\varepsilon + \zeta) - E_0 \leq K_R(\nu, \varepsilon) \leq K_h(\varepsilon - \zeta) + E_0. \quad (39)$$

More generally, a verified finite lower endpoint L_h^+ for allowance $\varepsilon + \zeta$ and a verified feasible finite upper endpoint U_h^- for allowance $\varepsilon - \zeta$ give the target interval

$$[\max(0, L_h^+ - E_0), U_h^- + E_0]. \quad (40)$$

The cellwise extension of the tightened-allowance policy is feasible for the original all-state contract. The endpoints use the original initial distribution through its exact cell weights.

PROOF: For any fixed Borel policy, subtract its two Bellman recursions, use the reward error, and apply the total-variation inequality to the surrogate continuation value. Backward induction gives $|J_t^p - \bar{J}_t^p| \leq A_t$. The maximum over finitely many actions is nonexpansive in the supremum norm, giving $|V_t - \bar{V}_t| \leq A_t$. Hence $|D_t^p - \bar{D}_t^p| \leq 2A_t \leq \zeta$. For costs use $0 \leq \bar{C}_{t+1}^p \leq K_* S_{T-t-1}$ to obtain $|C_t^p - \bar{C}_t^p| \leq E_t$.

Every feasible target policy is thus feasible for the surrogate at allowance $\varepsilon + \zeta$. By Lemma 3, its surrogate objective is at least $K_h(\varepsilon + \zeta)$. Taking infima and subtracting the objective error proves the lower inequality. Conversely, extend a finite policy feasible at $\varepsilon - \zeta$ to the cells. Its target regret is at most ε , and its target cost is at most its finite cost plus E_0 . Finite attainment gives the upper inequality, and the same argument applies to verified finite endpoints. *Q.E.D.*

COROLLARY 3—An explicit continuum stopping rule: *Suppose a sequence of certified surrogates has $a_r, a_k, a_g, a_P \rightarrow 0$, and let both finite endpoint computations have width at most η_h . For fixed T , $\beta < 1$, and $\varepsilon > 0$, the width in (40) is at most*

$$2E_0 + 2\eta_h + D_C^* \frac{2\zeta}{2\zeta + (1 - \beta)(\varepsilon - \zeta)}, \quad D_C^* = K_* \sum_{t=0}^{T-1} \beta^t S_{T-t}. \quad (41)$$

It therefore tends to zero as the surrogate errors and η_h tend to zero. Given a requested width, finite refinement can stop either when the directly computed interval meets it or when the displayed sufficient bound does.

PROOF: Start with an optimum for the finite model at allowance $\varepsilon + \zeta$. Relative to allowance $\varepsilon - \zeta$, it is 2ζ -infeasible. The exact-value repair argument of Theorem 4, with no witness or quantization error, gives a feasible policy at the smaller allowance after row movement at most

$$2\zeta / \{2\zeta + (1 - \beta)(\varepsilon - \zeta)\}.$$

The implementation-cost Lipschitz bound therefore bounds the difference between the two finite optima by the last term of (41). Each finite endpoint is within η_h of its optimum. Adding the two objective-transfer errors proves the result. Finite paired solvers from the preceding sections supply any fixed finite accuracy; the resulting continuum algorithm is constructive, although not uniformly inexpensive. *Q.E.D.*

8.3. Computing the primitive bounds from density regularity

For example, take $X = [0, 1]^d$, μ uniform, and a partition whose cells have diameter at most h . Suppose each target density $p_t^a(x, y)$ satisfies

$$|p_t^a(x, y) - p_t^a(x', y')| \leq L_x \|x - x'\| + L_y \|y - y'\|.$$

Average the density in the origin and destination cells. The resulting surrogate has the form (35), is nonnegative and normalized, and satisfies

$$a_P \leq \frac{1}{2}(L_x + L_y)h.$$

If rewards, costs, and the terminal reward are Lipschitz, cell averages give $a_r \leq L_r h$, $a_k \leq L_k h$, and $a_g \leq L_g h$. Verified integral enclosures and normalization corrections add their stated errors to these bounds. Thus all terms in (41) are calculable before claiming continuum accuracy. The number of cells still grows with state dimension.

A concrete action-dependent density is

$$p^a(x, y) = 1 + \vartheta_a(2x - 1)(2y - 1), \quad |\vartheta_a| \leq 1, \quad x, y \in [0, 1].$$

It is nonnegative and integrates to one. Its uniform-cell averages have rational transition probabilities whenever ϑ_a is rational, and the preceding bound gives $a_P \leq 2 \max_a |\vartheta_a|/N$ for N equal intervals. This example illustrates a directly checkable controlled density, not a reported high-dimensional performance result.

The theorem does not convert the earlier moving-atom uniform-initial-law computations into density computations. A point mass and a diffuse law have total-variation distance one. Those original intervals remain separate in the retained history. Likewise, model estimation error is not arithmetic error: applying (36) to estimated primitives requires justified simultaneous bounds covering the target law. The theorem specifies where those bounds would enter rather than assuming them from a numerical residual.

9. A DIFFUSE MODEL WITH DIRECTLY EXECUTABLE PAIRED CERTIFICATES

The general transfer theorem allows controlled densities. A reset subclass permits a direct integrated lower bound and avoids paying a conservative kernel-approximation error. Let $X = [0, 1]$, let ν and every transition kernel be uniform on $[0, 1]$, and take

$$r_t(x, a) = \gamma_a g_t(x), \quad 0 \leq \gamma_a \leq 1, \quad \max_a \gamma_a = 1,$$

with positive affine g_t and nonnegative affine implementation costs k_{ta} . The terminal reward is zero. Transitions have a continuous density and a diffuse initial law. They are independent of state and action, so this subclass exploits a reset structure rather than test the generic controlled-kernel search.

The operating advantage is $d_{ta}(x) = (1 - \gamma_a)g_t(x)$. Define cumulative regret moments

$$M_T = 0, \quad M_t = \int_0^1 \sum_a p_t(a | x) d_{ta}(x) dx + \beta M_{t+1}. \quad (42)$$

The continuation term in this equation is essential; M_t is not merely the one-period expected disadvantage. The original contract and objective become

$$\sum_a p_t(a | x) d_{ta}(x) + \beta M_{t+1} \leq \varepsilon \quad \text{for all } x, t, \quad (43)$$

$$C_0^p(\nu) = \sum_{t=0}^{T-1} \beta^t \int_0^1 \sum_a p_t(a | x) k_{ta}(x) dx. \quad (44)$$

Equations (42)–(44) are linear in the measurable probabilities and moments. They retain one common continuation moment at every date, rather than an independent choice of that moment for each restart.

PROPOSITION 3—Integrated lower certificate for all Borel policies: *Choose arbitrary non-negative bounded Borel multipliers $\lambda_t(x)$ and arbitrary real numbers q_t . Set $\Lambda_t = \int \lambda_t(x) dx$ and*

$$s_0 = -q_0, \quad s_t = -q_t + \beta(q_{t-1} + \Lambda_{t-1}) \quad (t \geq 1).$$

Then the following is a lower bound on the original unrestricted Borel-policy optimum:

$$\begin{aligned} \mathcal{L}(\lambda, q) = \max \bigg\{ & 0, \sum_t \int_0^1 \min_a \{ \beta^t k_{ta}(x) + (\lambda_t(x) + q_t) d_{ta}(x) \} dx \\ & - \varepsilon \sum_t \Lambda_t + \varepsilon \sum_t \min(0, s_t) \bigg\}. \end{aligned} \quad (45)$$

For stepwise constant λ_t and rational affine primitives, its integrals can be evaluated with finite rational operations.

PROOF: For a feasible policy, multiply (43) minus ε by $\lambda_t(x)$ and integrate. Add q_t times (42) written as $\int p_t d_t + \beta M_{t+1} - M_t = 0$. These additions do not increase the feasible objective. The coefficient of M_t is s_t . Every feasible moment lies in $[0, \varepsilon]$, so its contribution is at least $\varepsilon \min(0, s_t)$. Minimizing each policy row over the simplex gives the pointwise minimum in (45). This inequality holds for every feasible Borel policy before any cell restriction. Nonnegative costs permit clipping the lower bound at zero.

On a cell where λ_t is constant, each expression inside the minimum is affine. Partition the cell further at pairwise line intersections; one line minimizes on each resulting open interval. Its integral is the rational trapezoidal integral. Endpoint ties have zero Lebesgue measure and do not change the value. Q.E.D.

9.1. Constructing a globally feasible upper endpoint

Partition $[0, 1]$ into N equal intervals and use one policy row p_{tca} on each cell. Let $\bar{d}_{tca}$ and $\bar{k}_{tca}$ be their exact cell averages, and let $d_{tca}^{\max}$ be the maximum of the two endpoint disadvantages.

The following finite LP generates a proposal:

$$\begin{aligned}
 \min_{p, M} \quad & \sum_{t, c, a} \frac{\beta^t}{N} p_{tca} \bar{k}_{tca}, \\
 M_t = N^{-1} \sum_{c, a} p_{tca} \bar{d}_{tca} + \beta M_{t+1}, \\
 \sum_a p_{tca} d_{tca}^{\max} + \beta M_{t+1} \leq \varepsilon, \quad p_{tc} \in \Delta_m, \quad M_T = 0.
 \end{aligned} \tag{46}$$

The sums of actionwise maxima can be conservative when slopes differ. They guarantee the pointwise constraint without assuming a cellwise optimum of the unrestricted problem. The LP has $TNm + T$ variables and $2TN + T$ stated equality and inequality rows, before simple variable bounds.

Numerical probabilities are reconstructed as dyadic rationals and normalized. Process dates backward using the actual moment of the already reconstructed successor. The remaining allowance is $a_t = \varepsilon - \beta M_{t+1} > 0$. If a row's expected endpoint-maximum disadvantage exceeds a_t , mix toward an action with $\gamma_a = 1$ by the exact amount needed. This action has zero disadvantage. Round all other probabilities downward to a dyadic grid and transfer the remainder to the zero-disadvantage action. The row remains feasible. Recompute the exact integrated moment using (42), not the one-period integral alone. Since every row satisfies (43), integrating proves $M_t \leq \varepsilon$ and justifies the next step.

The resulting piecewise constant Borel policy provides an exactly integrated upper cost. The lower certificate uses the same continuous primitives but is not restricted to this policy class. A numerical LP status is neither necessary nor sufficient for accepting an endpoint: a failed numerical solve can supply a weaker proposal, and only the reconstructed inequalities determine validity.

9.2. Arithmetic, stopping, and approximation

Policy and multiplier proposals use 40-bit dyadic reconstruction. Each affine-envelope integral is evaluated rationally and rounded *downward* to 50 binary fractional bits before accumulation. This adds less than $TN2^{-50}$ of total downward error, prevents unnecessary denominator growth, and leaves the lower inequality valid. The upper endpoint is a rational integral of the deployed policy, without downward rounding. The independent readers repeat these conventions.

The operational stopping condition is the original continuous-model inequality $U - \mathcal{L} \leq 10^{-3}$. The same units apply to every primary model. Relative width and width normalized by the cost of the always-preferred action are secondary reports and do not change target status. A lower bound on implementation savings relative to that feasible reference is its exact cost minus U . A fixed administration fee is worth incurring only when it is smaller than a justified cost saving against the specified comparator; no dollar conversion is built into the certificate.

Refining the cells can be justified independently of a numerical solver's dual selection. Here $a_P = 0$ in Theorem 6; affine reward and cost approximation errors are explicitly of order N^{-1} . The cell model has action-independent transitions, so its common-policy moment formulation is a finite LP. Solving the two corrected-allowance finite programs and using Corollary 3 gives a complete paired approximation algorithm for this subclass. The direct integrated dual often gives a substantially narrower interval than that sufficient transfer bound. We accept the direct interval only through its own certificate; we do not assume that every sequence of floating LP marginals must converge.

10. NUMERICAL EVIDENCE AND DECISION ACCURACY

The experiments distinguish the original finite primary cohort from a new diffuse-law cohort. A closure means that the independently reconstructed upper minus lower endpoint does not exceed 10^{-3} . A root closure, a zero-cost closure, a branching closure, and a smaller unfinished interval are different outcomes. No failed case is replaced by a more favorable draw.

10.1. Which assumptions have been executed?

Table I separates mathematical coverage from computation. An arithmetic reader verifies an instance of a specified certificate. It does not verify the theorem that connects the certificate specification to Borel policies, nor establish that designed primitives describe an actual institution.

TABLE I
ASSUMPTIONS, CONCLUSIONS, AND EXECUTED EVIDENCE

Object	Mathematical condition and conclusion	Executed evidence
Common-policy finite model	Finite actions, rational primitives; complete covers and policy repair give paired bounds	All 12 historical primary models; 48 new capped runs, including failures
Price optimum	All positive-support masks; feasible occupation upper witnesses	All eight historical unfinished rows; uncapped and four capped portfolios
Finite atomic continuum	Finite initial support and finite all-action forward closure	Two designed fleets; 13 and 76 active decision nodes
Density approximation	Certified uniform primitive errors and total-variation kernel bound	General transfer theorem; no claim of a general-density performance benchmark
Uniform-reset continuum	Uniform initial law and reset density; affine primitives	39 source-frozen models, two proposal backends; whole-cell upper checks and Borel lower bounds

10.2. A source-frozen diffuse-law study

The protocol and hashes of the constructors, primary drivers, and Python readers were committed before any primary seed was generated. The freeze commit is 62585130b01f1b3479b5fc9f8ee0ee927509633a. This is an auditable within-project freeze, not external preregistration. Development used seed 9001, constant-affine analytic examples, and one historical warranty model, not a primary seed.

The baseline has $T = 8$, $m = 3$, $\beta = 0.9$, $\varepsilon = 0.05$, and spatial-slope parameter 0.4. Table II varies one factor at a time. Each configuration uses seeds 401, 503, and 607. The generated functions g_t are strictly positive affine functions; costs are nonnegative affine functions. The action productivity grid includes 0 and 1. Every transition draws a uniform state independently of the preceding state and action, and the initial law is uniform. This is a genuinely diffuse, non-finite-support target with exploitable exogenous-reset structure, not a difficult action-dependent transition benchmark.

For each model, dual-simplex and interior-point proposals use the same grid sequence 16, 64, 256, 1024 and a 16-second soft LP limit at each level. The policy and multiplier fields are reconstructed rationally. The driver stops only after its independent reader accepts both

endpoints and the width test. All intermediate levels, LP statuses, final policies, multipliers, exact endpoints, proof hashes, and work measurements are retained. The two backends belong to the same HiGHS optimization stack; they are not independently authored global solvers.

TABLE II
DIFFUSE-LAW CERTIFICATION: THREE SEEDS PER CONFIGURATION

Factor	DS hits	IPM hits	Max N	Max width	DS sec.	IPM sec.	Max MiB
base	3/3	3/3	256	0.000971	0.911	0.937	88.9
T4	3/3	3/3	256	0.000823	0.660	0.663	81.6
T16	3/3	3/3	256	0.000876	4.645	5.560	102.6
T32	3/3	3/3	256	0.000877	8.874	9.764	129.6
m2	3/3	3/3	256	0.000512	1.874	1.918	85.4
m5	3/3	3/3	256	0.000866	3.929	4.134	95.0
beta50	3/3	3/3	256	0.00079	0.924	0.931	89.9
beta97	3/3	3/3	256	0.000798	2.556	2.717	89.8
beta99	3/3	3/3	256	0.0008	2.532	2.767	91.2
eps02	3/3	3/3	256	0.000971	0.908	0.935	88.9
eps15	3/3	3/3	256	0.000971	0.907	0.931	88.8
slope0	3/3	3/3	256	0.000898	0.898	0.927	88.2
slope80	3/3	3/3	256	0.000734	2.560	2.884	89.0

DS and IPM columns report median all-in seconds across three separate worker processes. Width is an absolute implementation-cost interval. T4/T16/T32 vary horizon; m2/m5 vary actions; beta50/beta97/beta99 vary discounting; eps02/eps15 vary allowance; slope0/slope80 vary the spatial slope. The baseline remains fixed in the other coordinates.

Raw results give every exact endpoint, not only these aggregates.

The publication run records 78 target hits across 78 model–backend pairs, with 200 grid certificates. The largest final LP has 24608 variables. Aggregate worker all-in time is 188.14 seconds and the largest process high-water memory is 129.6 MiB. These are measured resources on the recorded environment, not complexity constants. A worker’s timer includes imports within the worker, construction, serialization, and its Python check; interpreter startup before that timer and the separate cross-language audit are not charged to this number. They are included separately in reproduction logs. Each backend’s memory comes from a new process.

A second reader uses JavaScript BigInt rational arithmetic and imports neither Python nor the constructor. It checks the final dual-simplex proof for seed 401 in each of the thirteen configurations, including all-state endpoint inequalities and the complete cumulative-regret recursion. All thirteen checks are retained. Ten analytic and adversarial contract tests include known optimal values at four horizons and rejection of altered moments, inflated lower bounds, invalid prices, altered models, invalid normalization, and whole-cell infeasibility. This is cross-language implementation evidence within the project, not independent scientific replication by another research group.

10.3. Finite-model structure and its computational limits

The budget-coordinate method changes the search object rather than merely adding another price. The new comparison reruns both methods from raw R48 primitives, without inherited incumbents, for every primary model at two and eight seconds of declared construction work. Table III reports all widths. The node limit is 511. An exact operation already in progress is completed rather than interrupted, and its time is included. Thus these are common *soft construction caps*, not equal elapsed-time measurements.

TABLE III
HISTORICAL FINITE MODELS: NEW CAPPED CONSTRUCTIONS

Case	Price, 2 sec.	Budget, 2 sec.	Price, 8 sec.	Budget, 8 sec.	d_B/d_P
inventory0	0.000892	0.00528	0.000892	0.00328	9/36
inventory1	0.0249	0.0786	0.0216	0.0672	35/120
inventory2	0.0517	0.0494	0.017	0.0476	88/384
queue0	9.3e-09	9.3e-09	9.3e-09	9.3e-09	9/36
queue1	0.00517	0.0358	0.00301	0.0357	35/120
queue2	0.0151	0.0349	0.014	0.0349	88/384
ties0	0.0175	0.00689	0.00909	0.00543	9/36
ties1	0	0	0	0	35/120
ties2	0.0572	0.0525	0.0524	0.0525	88/384
warranty0	0.000987	0.00307	0.000987	0.00307	9/36
warranty1	0.0156	0.0303	0.0139	0.0303	35/120
warranty2	0.0198	0.0286	0.016	0.0286	88/384

$d_B = n(T - 1)$ and $d_P = nTm$. All entries are certified widths, not solver-reported gaps. Each retained result reports actual total time, root/tree data, peak memory, and proof size. The price driver and budget driver have different frozen local-proposal allocations; this is a comparison of two complete constructors, not an isolated causal ablation of the branching coordinate alone.

The budget constructor achieves 2/12 and 2/12 closures; the price constructor achieves 4/12 and 4/12. Reducing the coordinate count is therefore not sufficient evidence of faster certification. The completion theorem remains useful because it identifies the consistency variable and supplies a quantitative error modulus. It does not remove the need for effective node bounds, incumbents, and branching in general controlled dynamics. No median or count discards an unfinished run. The supplement and machine-readable work curves expose the full comparison.

10.4. How much can further price search achieve?

The retrospective audit in Table IV concerns the *ideal uncapped price relaxation*. It computes a lower field and an upper bound from feasible occupation witnesses for every initial-support mask, using policy-cost upper bounds to prune masks when justified. A mask is never dropped merely because the state count is large. A verified upper bound on each mask also bounds the clipped maximum. Zero-flow edges impose no ratio restriction on their budget allocation, as both allocation masses are zero.

In 6 of eight rows the price optimum is bracketed to less than 10^{-8} . The other two rows are retained with their wider upper witnesses. The audit also tries all support masks at caps 256, 1024, 4096, and 16384. For every computed capped portfolio, the uncapped upper endpoint minus the replayed lower endpoint is a valid bound on the combined loss from the cap, mask scheduling, numerical proposals, and unfinished optimization. These upper losses and every attempted-mask status appear in the supplement. The cap 4096 has no institutional interpretation.

The audit reconstructs the exact finite operating value from the raw finite model. It is a diagnostic of the price family, separate from the witness-only primary constructors. It is not passed to their price or policy generation. The upper occupation point may allocate different regret shares on incoming edges; this is precisely why a tight price interval does not certify the common-policy optimum. The distinction permits a positive diagnostic: a large remaining common-policy interval in a row with a tightly resolved price optimum cannot be attributed solely to failure to maximize that price family.

TABLE IV
CERTIFIED INTERVALS FOR THE UNCAPPED RESTART-PRICE OPTIMUM

Case	Masks	Occ. witnesses	Pruned	Lower	Upper	Width
inventory1	31	3	28	4.960238	4.960239	7.93e-15
inventory2	3	2	1	10.196728	10.196729	8.84e-14
queue1	31	3	28	7.388772	7.388773	1.24e-13
queue2	3	2	1	10.617117	10.726004	0.109
ties0	7	3	4	1.049693	1.049694	6.81e-15
ties2	3	2	1	6.082702	6.082703	2.92e-09
warranty1	31	4	27	8.205716	8.414980	0.209
warranty2	3	2	1	8.897888	8.897889	4.71e-12

Occupation witnesses are exactly feasible rational points, not claims of exact LP optimality. The interval brackets the price-family optimum, not the original common-policy optimum. Displayed decimals summarize exact rational endpoints retained in the proof files.

10.5. Historical evidence is retained, not relabeled

Table V reproduces the exact R48 primary records. Direct and price formulations each close four of twelve rows; prices add zero closures. The three positive-cost successes W0, I0, and Q0 are at $(n, T, m) = (3, 4, 3)$; T1 is a zero-cost closure. Q0 closes at the root, while W0 and I0 require branching. No positive-cost medium or large row closes. Aggregate recorded work was approximately 286.26 seconds for direct construction and 340.52 for price construction. The joint variation of dimensions, discounting, and allowance in that cohort is not used to infer one-factor scaling.

TABLE V
PRESERVED R48 PRIMARY COHORT

Case	Price upper	Direct width	Price width	Relative %	Nodes	Stop
W0	3.110700	0.000995	0.000987	0.0317	5	target
W1	8.236030	0.128	0.0126	0.1535	169	time_cap
W2	8.926491	0.0447	0.0111	0.1243	27	time_cap
I0	1.982232	0.000997	0.000892	0.0450	9	target
I1	5.044754	0.0509	0.0123	0.2440	203	time_cap
I2	10.244317	0.0718	0.0109	0.1061	23	time_cap
Q0	0.230546	9.3e-09	9.3e-09	0.0000	1	target
Q1	7.424602	0.0238	0.00237	0.0320	191	time_cap
Q2	10.652032	0.0506	0.0117	0.1102	29	time_cap
T0	1.074850	0.151	0.00855	0.7953	255	node_cap
T1	0.000000	0	0	0.0000	1	target
T2	6.135079	0.95	0.0524	0.8537	39	time_cap

The entire R48 article, including its longer numerical discussion and examples, is reconstructed with the missing generated tables and included in the historical part of the supplement. Its text is identified by reviewed commit and is not silently rewritten as current evidence. Earlier revision manuscripts, proofs, and reports remain in the repository. In particular, all thirty positive-cost uniform-initial intervals for the moving-atom benchmark remain historical unfinished intervals.

TABLE VI
PRESERVED FINITE-OBSERVATION FLEETS

Case	Price upper	Direct width	Price width	Relative %	Nodes	Stop
F0	1.269602	2.28e-09	2.28e-09	0.0000	1	target
F1	2.868705	0.000367	8.27e-09	0.0000	1	target

Designed two-point initial law; horizons two and three, with 13 and 76 active decision nodes. These are not observations from fleet data and are not the uniform-initial moving-atom model. Exact forward closure has zero spatial approximation error but a worst-case exponentially growing graph.

10.6. Economic interpretation and conclusion

Let C^{op} be the exact implementation cost of the operating-best policy in a reset model. The quantity $C^{\text{op}} - U$ is a certified implementation saving achievable without breaking any restart's operating promise. For a fixed administration fee f , a certified cost advantage follows when $C^{\text{op}} - U > f$; no such advantage is possible when $C^{\text{op}} - L < f$. The interval width bounds the additional implementation expense of the deployed policy relative to the least feasible expense. The supplement reports $(U - L)/U$, $(U - L)/C^{\text{op}}$, and $C^{\text{op}} - U$ for every primary reset model. These comparisons are invariant to a common positive scaling of costs, fees, budgets, and absolute accuracy. They are conditional economic decisions, not estimates of welfare or market prices.

The paper's target remains certified minimum-cost revision under one common policy. The new results give an explicit price-optimization loss bound, a smaller continuation-budget cover with a repair modulus, and paired approximation for density kernels. The executed reset problem supplies diffuse initial laws, continuous kernels, global feasibility, and Borel-valid lower endpoints within a single solver. General finite positive-cost scalability, action-dependent diffuse-kernel performance, expensive learned operating witnesses, statistical model uncertainty, and a calibrated economic application are not inferred from that structured success. These distinctions preserve the research object while locating exactly which parts are theorem, executed certificate, and further empirical validation.

APPENDIX A: ADDITIONAL ERROR AND DUAL DETAILS

A.1. Operating-witness sensitivity of a price field

Suppose u is reconstructed with exact disadvantage d , and $\underline{u}$ with (18), using the same nonnegative field. The inequalities

$$0 \leq d_{tia} - \underline{d}_{tia} \leq w_t + \beta w_{t+1}$$

follow directly from the operating enclosure (including the clipping at zero). Backward minimum stability yields

$$0 \leq u_{ti} - \underline{u}_{ti} \leq \Gamma_t, \quad \Gamma_t = \sum_{s=t}^{T-1} \beta^{s-t} \|\lambda_s\|_{\infty} (w_s + \beta w_{s+1}).$$

The positive-part map is 1-Lipschitz, so the same Γ_0 bounds the change in the displayed clipped lower endpoint. Any directed approximation of the price expectation adds its certified one-step error to this recursion. In the executed model the expectation is an exact rational finite sum. This is a perturbation result for a *fixed* field; the algorithm's cap-based rule is imposed before generating the field, so it does not presuppose stability of an exact-model price optimizer.

A.2. Clipping can change the optimizing price

The mask maximization is not merely a different display of the full-mask optimum. Modify the two-period example in Theorem 2 as follows. Keep $\beta = \varepsilon = 1/2$ and deterministic transitions to state 1. At date 0, rewards of G, B are 0, -1 , while costs are $(0, 0)$ in state 1 and $(3, 0)$ in state 2. At date 1, rewards of G, B are 0, $-1/4$ and costs are $(1, 0)$ in both states. Let $\nu = (3/4, 1/4)$ and $g = 0$. The operating value is zero.

Let l be the price at the reached terminal decision state. Its potential is

$$u_1(l) = \min\{1 - l/2, -l/4\}.$$

For this fixed l , the best root potentials use prices 0 and 3 and equal $u_{01} = u_1(l)/2$ and $u_{02} = 3/2 + \min(3, l)/4 + u_1(l)/2$. The first claim follows from the zero-cost good action; the second follows by checking the positive slope before root price 3 and the negative slope thereafter. Thus the optimized full-mask objective is

$$3/8 + \min(3, l)/16 + u_1(l)/2,$$

whose maximum $3/8$ occurs at $l = 0$. Its clipped value there is also $3/8$. But $l = 3$ gives potentials $(-3/8, 15/8)$ and clipped value $15/32 > 3/8$. The latter is the actual common-policy optimum: take the terminal bad action with probability 1, use the root bad action with probability $3/8$ in state 2, and use the good action in state 1. Its operating shortfalls obey all restart bounds, its objective is $15/32$, and the displayed certificate matches it. Hence optimizing the un-clipped objective can select a strictly inferior clipped certificate. Enumerating the relevant initial masks fixes this mismatch without altering the validity of any field.

A.3. A quantitative form of repaired Borel upper density

The inherited upper-density result can be stated with a finite error modulus, although not with a universal computable partition-size rate. Fix a globally feasible policy p and define finite dominating measures

$$\mu_0 = \nu, \quad \mu_{t+1} = \sum_{a=1}^m \mu_t P_t^a.$$

Their total masses are m^t . For any Borel proposal q , let $e_t(x) = \text{TV}(q_t(\cdot | x), p_t(\cdot | x))$ and $E_t = \int e_t d\mu_t$. Put $M = 2(RS_T + G)$, $s = (1 - \beta)\varepsilon$, $c = 1 + M/s$, and

$$A_t = cE_t + \frac{Mc\beta}{s} \sum_{v=t+1}^{T-1} (\beta c)^{v-t-1} E_v,$$

$$\mathcal{B}(E) = K \sum_{t=0}^{T-1} \beta^t S_{T-t} A_t.$$

Then exact continuation-best repair of q gives a globally feasible $\hat{p}$ with

$$|C_0^{\hat{p}}(\nu) - C_0^p(\nu)| \leq \mathcal{B}(E). \quad (47)$$

To verify this, write $a_t = \text{TV}(\hat{p}_t, p_t)$ and $\Delta_t = |J_t^{\hat{p}} - J_t^p|$. The threshold deficit before repair is at most $Me_t + \beta P_t^p \Delta_{t+1}$ because p is feasible. The best repaired continuation has margin at

least s , so the repair mass is at most this deficit divided by s . Therefore

$$a_t \leq ce_t + (\beta/s)P_t^p \Delta_{t+1}, \quad \Delta_t \leq Mce_t + \beta c P_t^p \Delta_{t+1}.$$

Integrating and using $\mu_t P_t^p \leq \mu_{t+1}$ proves $\int a_t d\mu_t \leq A_t$. The implementation performance-difference expansion and the range bound KS_{T-t} then give (47).

Rational simple proposals can approximate p in these finitely many $L_1(\mu_t)$ norms; the displayed expression gives value convergence after repair. But the dominating masses and factors c can grow rapidly, arbitrary Borel integrals need not be effectively evaluable, and a repaired simple proposal need not itself remain simple. Thus (47) is not a two-sided computable scheme for a diffuse initial law. The finite-observation theorem in the main text obtains such a scheme by exact closure, not by suppressing these difficulties.

A.4. Exact finite products and common budgets

For $p \in [l, u]$, $v \in [a, b]$, each product $z = pv$ obeys the sign-safe inequalities

$$\begin{aligned} z &\geq ap + lv - la, & z &\geq bp + uv - ub, \\ z &\leq ap + uv - ua, & z &\leq bp + lv - lb. \end{aligned}$$

The displayed four McCormick inequalities, not any sign shortcut, are reconstructed by the reader. Signed $B = \underline{V} - J$ variables are allowed. For a probability simplex, $\sum_a p_a v = v$ supplies an exact product conservation identity. These constraints preserve one continuation variable per successor. The occupation completion (11) is an alternative exact formulation, while the implemented probability-box search enforces the same compatibility through the original Bellman equations. The occupancy LP itself is not falsely described as complete for K_R .

APPENDIX B: ORGANIZATION OF RETAINED MATERIAL

The numerical supplement contains all 28 new rows, all recorded search trajectories, oracle conditioning, work components, the complete new source/test contract, and a compact assumption matrix. It also reproduces the complete R47 main article and supplement, which retain the strict-gap quadratic analysis, earlier finite searches, exact aggregation, reset transfer, observable fibers, original uniform-initial-law certificates, and nonlinear examples. A separate historical volume preserves the earlier R47 historical material. No earlier source, result, proof object, or review is deleted or modified. The preservation manifest checks that statement against the immutable reviewed baseline; preservation itself is not evidence for a new theorem or numerical target hit.

REFERENCES

- Altman, E. (1999): *Constrained Markov Decision Processes*. Chapman and Hall/CRC. [2]
Brown, D. B., and J. E. Smith (2014): “Information Relaxations, Duality, and Convex Stochastic Dynamic Programs,” *Operations Research*, 62(6), 1394–1415. doi:10.1287/opre.2014.1322. [2]
Brown, D. B., J. E. Smith, and P. Sun (2010): “Information Relaxations and Duality in Stochastic Dynamic Programs,” *Operations Research*, 58(4, part 1), 785–801. doi:10.1287/opre.1090.0796. [2]
Brumm, J., and S. Scheidegger (2017): “Using Adaptive Sparse Grids to Solve High-Dimensional Dynamic Models,” *Econometrica*, 85(5), 1575–1612. [3]
de Farias, D. P., and B. Van Roy (2003): “The Linear Programming Approach to Approximate Dynamic Programming,” *Operations Research*, 51(6), 850–865. doi:10.1287/opre.51.6.850.24925. [2]

- Dufour, F., and T. Prieto-Rumeau (2013): “Finite Linear Programming Approximations of Constrained Discounted Markov Decision Processes,” *SIAM Journal on Control and Optimization*, 51(2), 1298–1324. doi:10.1137/120867925. [2]
- Hettich, R., and K. O. Kortanek (1993): “Semi-Infinite Programming: Theory, Methods, and Applications,” *SIAM Review*, 35(3), 380–429. doi:10.1137/1035089. [3]
- Iyengar, G. N. (2005): “Robust Dynamic Programming,” *Mathematics of Operations Research*, 30(2), 257–280. doi:10.1287/moor.1040.0129. [2]
- Judd, K. L. (1998): *Numerical Methods in Economics*. MIT Press. [3]
- Laroche, R., P. Trichelair, and R. Tachet des Combes (2019): “Safe Policy Improvement with Baseline Bootstrapping,” *Proceedings of Machine Learning Research*, 97, 3652–3661. [3]
- Lasserre, J. B. (2001): “Global Optimization with Polynomials and the Problem of Moments,” *SIAM Journal on Optimization*, 11(3), 796–817. doi:10.1137/S1052623400366802. [3]
- McCormick, G. P. (1976): “Computability of Global Solutions to Factorable Nonconvex Programs: Part I—Convex Underestimating Problems,” *Mathematical Programming*, 10, 147–175. doi:10.1007/BF01580665. [3]
- Nilim, A., and L. El Ghaoui (2005): “Robust Control of Markov Decision Processes with Uncertain Transition Matrices,” *Operations Research*, 53(5), 780–798. doi:10.1287/opre.1050.0216. [2]
- Rump, S. M. (2010): “Verification Methods: Rigorous Results Using Floating-Point Arithmetic,” *Acta Numerica*, 19, 287–449. [3]
- Rust, J. (1987): “Optimal Replacement of GMC Bus Engines: An Empirical Model of Harold Zurcher,” *Econometrica*, 55, 999–1033. doi:10.2307/1911259. [3]
- Saldi, N. (2018): “Finite-State Approximations to Discounted and Average Cost Constrained Markov Decision Processes,” arXiv:1807.02994. [2]